3.1 Number

3.1.1 Structure and calculation

N1

Basic foundation content	Additional foundation content	Higher content only
order positive and negative integers, decimals and fractions use the symbols =, ≠, <, >, ≤, ≥		

Notes: including use of a number line. See also A22

N2

Basic foundation content	Additional foundation content	Higher content only
apply the four operations, including formal written methods, to integers, decimals and simple fractions (proper and improper), and mixed numbers – all both positive and negative understand and use place value (eg when working with very large or very small numbers, and when calculating with decimals)		

Notes: including questions set in context.

Knowledge and understanding of terms used in household finance, for example profit, loss, cost price, selling price, debit, credit, balance, income tax, VAT and interest rate. See also R9

N3

Basic foundation content	Additional foundation content	Higher content only
recognise and use relationships between operations, including inverse operations (eg cancellation to simplify calculations and expressions) use conventional notation for priority of operations, including brackets, powers, roots and reciprocals		

N4

Basic foundation content	Additional foundation content	Higher content only
use the concepts and vocabulary of prime numbers, factors (divisors), multiples, common factors, common multiples, highest common factor, lowest common multiple, prime factorisation, including using product notation and the unique factorisation theorem		

Notes: prime factor decomposition including product of prime factors written in index form.

N5

Basic foundation content	Additional foundation content	Higher content only
apply systematic listing strategies		including use of the product rule for counting

Notes: including using lists, tables and diagrams.

N6

Basic foundation content	Additional foundation content	Higher content only
use positive integer powers and associated real roots (square, cube and higher), recognise powers of 2, 3, 4, 5		estimate powers and roots of any given positive number

Notes: including square numbers up to 15 × 15

Students should know that $1000 = 10^3$ and 1 million $= 10^6$

N7

Basic foundation content	Additional foundation content	Higher content only
	calculate with roots, and with integer indices	calculate with fractional indices

N8

Basic foundation content	Additional foundation content	Higher content only
calculate exactly with fractions	calculate exactly with multiples of π	calculate exactly with surds simplify surd expressions involving squares (eg $\sqrt{12} = \sqrt{4 \times 3} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$) and rationalise denominators

Notes: see also G17 and G18

N9

Basic foundation content	Additional foundation content	Higher content only
calculate with and interpret standard form $A \times 10^n$, where $1 \leq A < 10$ and n is an integer		

Notes: with and without a calculator.

Interpret calculator displays.